ZCAS University

CCS1601 INTERNET OF THINGS FUNDAMENTALS

FINAL EXAM

18TH JUNE 2025

Time: 14:00Hrs – 17:00 Hours

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Section B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

Smart Home Greenhouse System

A team of university students is developing a smart greenhouse system using IoT technologies to automate plant care. Their system is powered by an **Arduino Uno** connected to various sensors and actuators on a **solderless breadboard** for prototyping.

The setup includes:

- **Soil moisture sensor**, **DHT22 temperature/humidity sensor**
- **Water pump and fan** connected via **transistor switches** as **actuators**
- A **relay module** to control AC grow lights
- An **ESP8266 Wi-Fi module** for internet connectivity
- All components are interfaced with an **Arduino Uno**, with power and control circuits prototyped on a **breadboard**
- Data is sent to a **cloud dashboard** for real-time monitoring and remote actuation

The students face several challenges in understanding the interaction of electronic components, power supply management, programming logic for sensors and actuators, and cloud integration.

You have been hired as an IOT Specialist to help them complete the task.

a) Explain the role of a microcomputer (e.g., Arduino Uno) in this IoT project.

(4 marks)

b) Discuss why Arduino is suitable for prototyping in educational environments.

(4 marks)

(c) The system uses a solderless breadboard during prototyping.

i) Illustrate how a breadboard works, highlighting the internal connections.

ii) Discuss two advantages and two limitations of using a solderless breadboard in circuit design.

(6 marks)

d)The water pump and fan are connected via transistors as switching devices.

SECTION B: Attempt any THREE questions in this section.

Question 2

a) Shown below is a code that prints a list of fruits. Write a code using a FOR LOOP to INDEX the list of fruits. Show your output

```
fruits = ["apple", "orange", "tomato", "banana"]  # a list of strings
print(type(fruits))
print(fruits)
```

```
<class 'list'>
['apple', 'orange', 'tomato', 'banana']
```

(6 marks)

b) Write Python code that executes TEN numerical values and squares them. (8 marks)

c) Write a program of your choice using a WHILE LOOP. (6 marks)

(20 marks)

Question 3

Electronics is a field of study focused on the control of electricity and the physical components and circuits that help direct electrical energy.

(a) Draw a basic circuit diagram (schematic) showing the current, resistor, and LED.

(5 marks)

(b) Define cloud computing and describe the user benefits of IaaS, PaaS, mPaaS, SaaS

(10 marks)

(c) Explain what the Rasberry Pi and Arduino board is used for in IoT (5 marks)

(20marks)

Question 4

Functions allow us to package functionality in a nice and readable way, reuse it without writing it again, and make code modular and readable, Rule of thumb - if you are planning on using very similar code more than once, it may be worthwhile writing it as a reusable function.

(a) Write a Python program using the following functions showing indentation.

i. def (5 marks)

i) Explain how an **NPN transistor** can be used as an actuator switch in this system.

ii) Draw a simple circuit diagram showing an Arduino digital output controlling a transistor to switch a DC motor.

(6 marks)

(e) The greenhouse system needs to respond to sensor input and control actuators accordingly.

i) Write a **pseudo-code** or code snippet to read from a **soil moisture sensor** and activate a water pump when the soil is dry.

ii) Briefly explain the purpose of using **digitalWrite()** and **analogRead()** functions in Arduino programming.

(7 marks)

f) Discuss how the system uses an **ESP8266 Wi-Fi module** for IoT functionality:

i) Describe the purpose of integrating Wi-Fi in the system.

ii) List and explain two key challenges in using Wi-Fi modules with microcontrollers.

(6 marks)

g) Sensors and actuators must be powered correctly to avoid damage.

i) Explain why it is important to manage voltage and current levels in circuits involving microcontrollers and actuators.

ii) Describe how a relay module can safely control high-voltage devices (e.g., AC grow lights) using an Arduino.

(6 marks)

(Total: 40 marks)

ii. if else (5 marks)

iii. break (5 marks)

(b) Write a Python program the defines a dictionary using the type (days of the week) (5 marks)

(20 marks)

Question 5

Cloud customers have access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort. Discuss the following.

a) The Fog Computing Model (5marks)

b) Big Data and Security concerns in IoT (5 marks

c) Data Transmission (5 marks)

d) Cloud Computing (5 marks)

(20 marks)

END OF EXAM